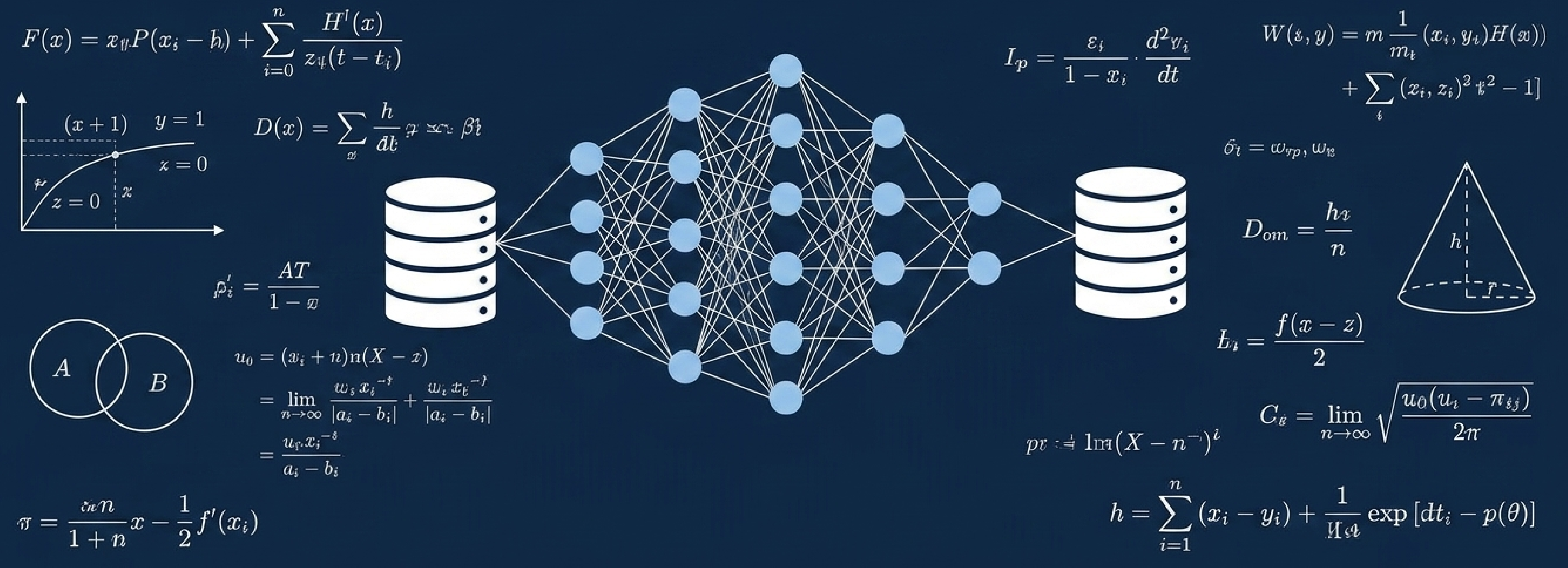
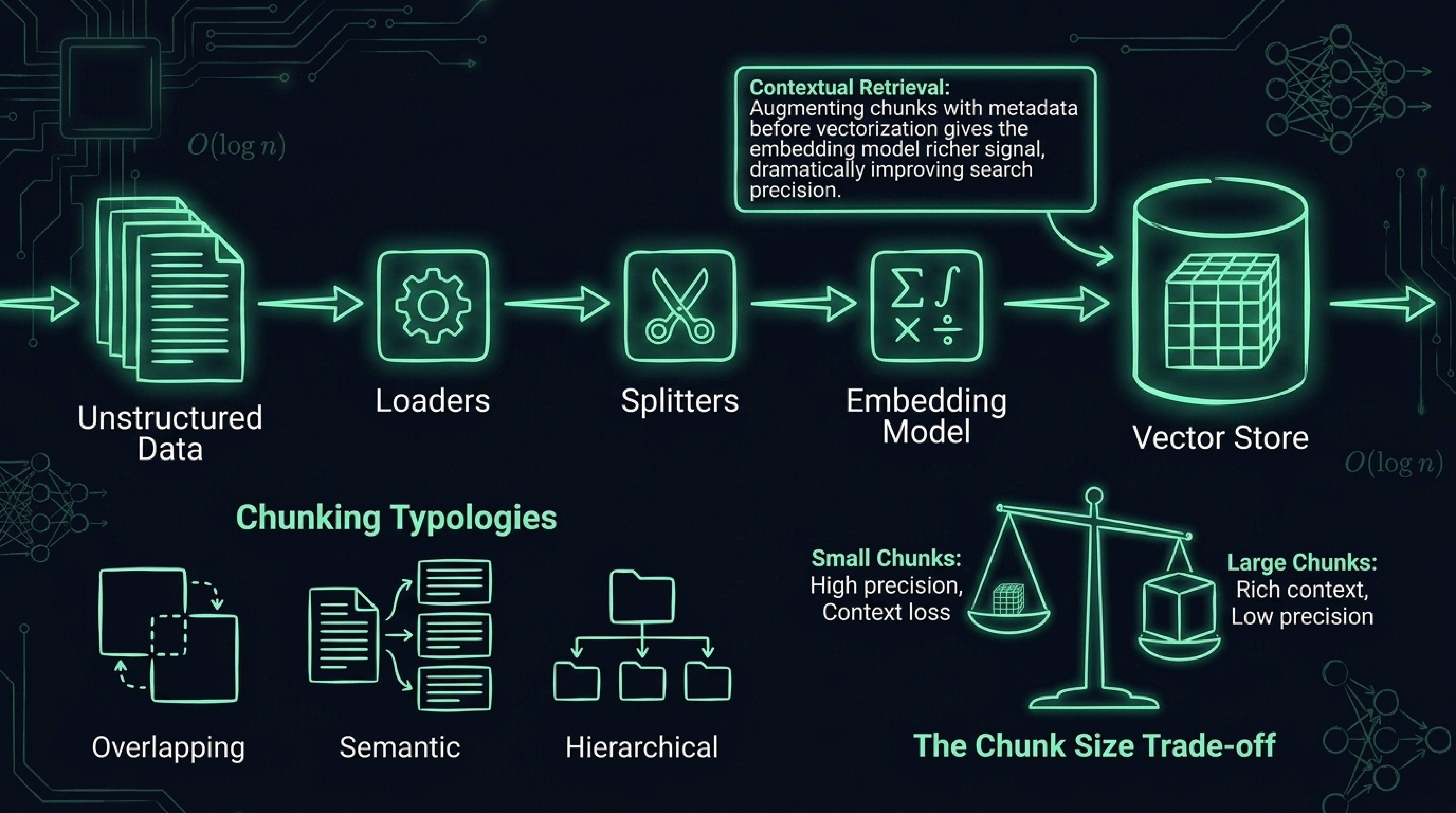
Retrieval-Augmented Generation (RAG)

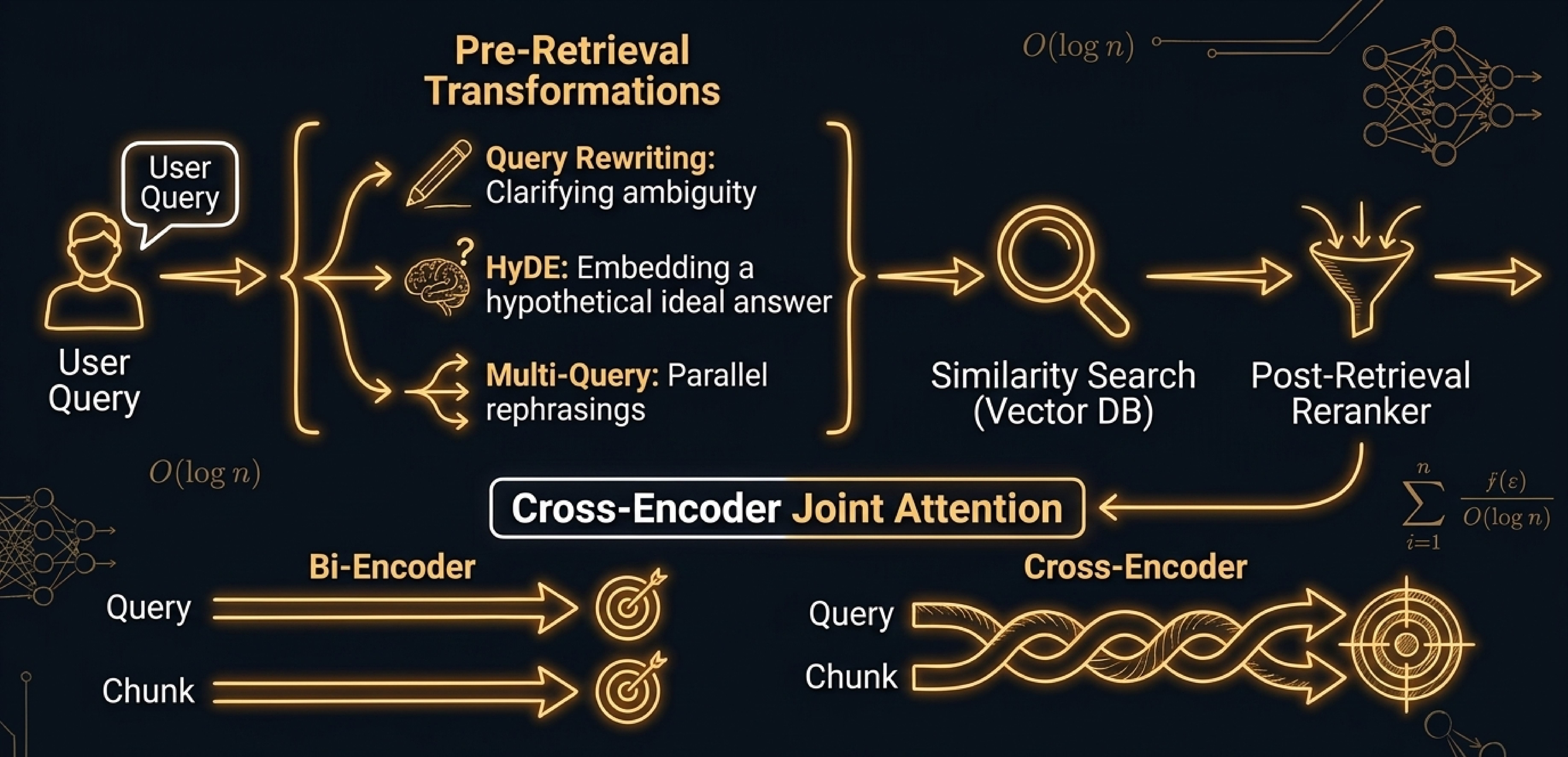
By Michaël BETTAN



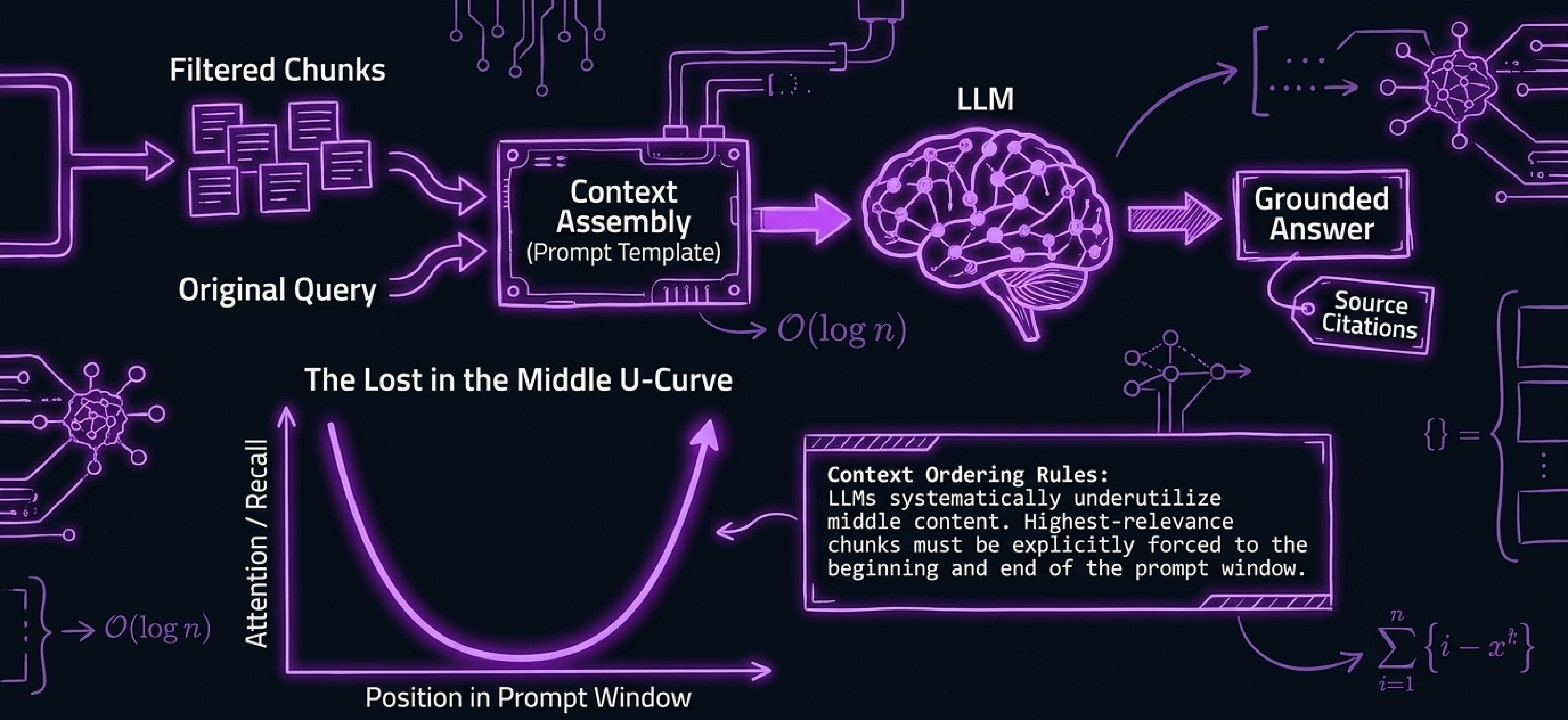
	RAG	Fine-Tuning	Long Context Windows
Primary Purpose	**Fact** injection	**Form, style, jargon**	**Brute-force** ingestion
Hallucination Risk	**Low risk** (falsifiable)	**High risk** on **facts**	**Moderate risk** (Lost-in-the-Middle bias)
Token Cost	**Highly efficient** (dense payloads)	**Training cost** overhead	**Financially unscalable** per query
Latency / TTFT	**Sub-second**	**Fast**	**Severe parsing** overhead
Data Freshness	**Real-time updates**	**Frozen** at **training cutoff**	**Dynamic** but **slow**

RAG is the **architecturally correct** solution for **factual grounding** at **inference time**—supplying the **exact facts** while **fine-tuning** ensures the **correct format**.

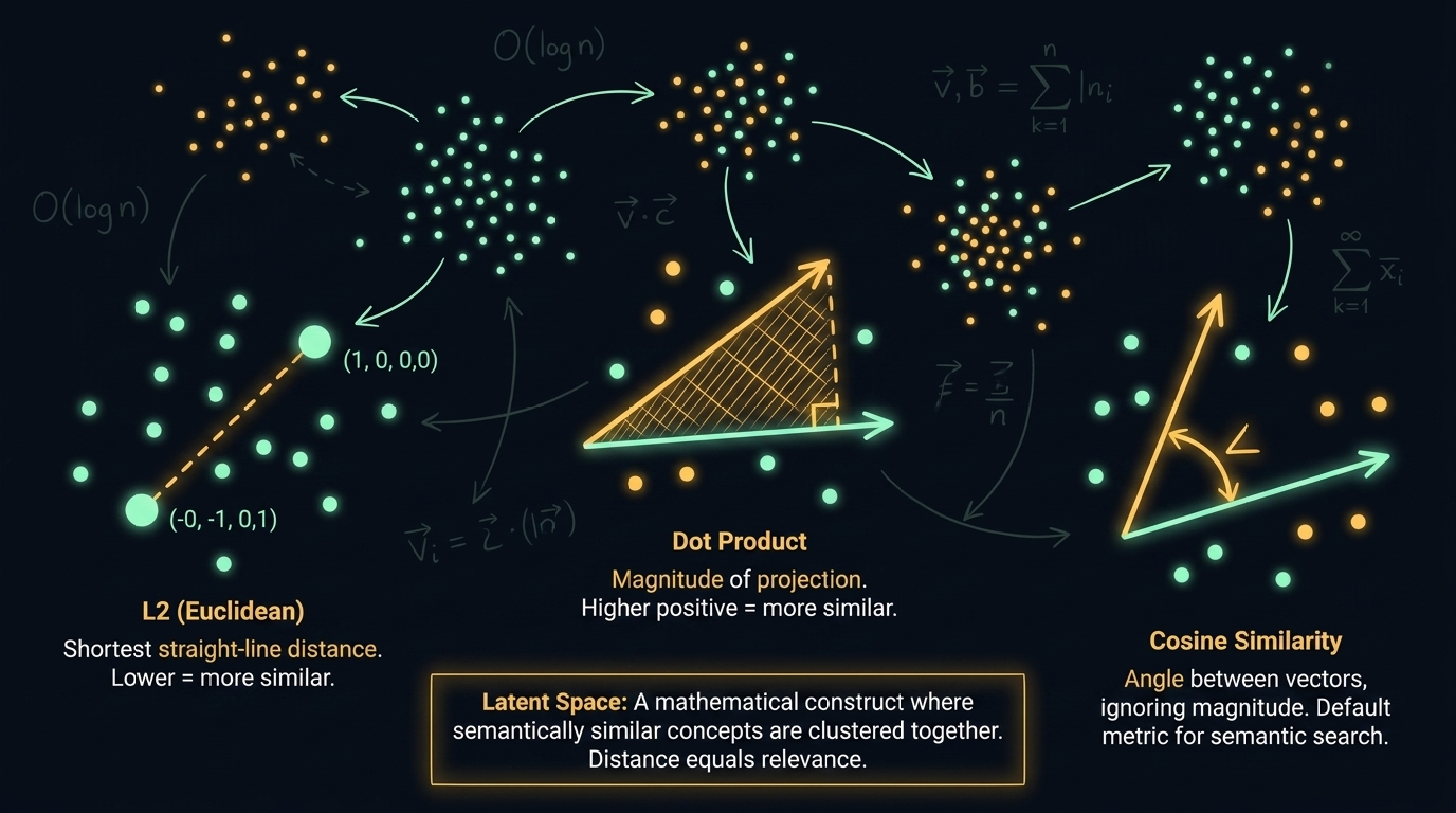

Contextual Retrieval:
Augmenting chunks with metadata before vectorization gives the embedding model richer signal, dramatically improving search precision.
$O(\log n)$
Unstructured Data
Loaders
Splitters
Embedding Model
Vector Store
$O(\log n)$
Chunking Typologies
Overlapping
Semantic
Hierarchical
Small Chunks:
High precision,
Context loss
Large Chunks:
Rich context,
Low precision
The Chunk Size Trade-off

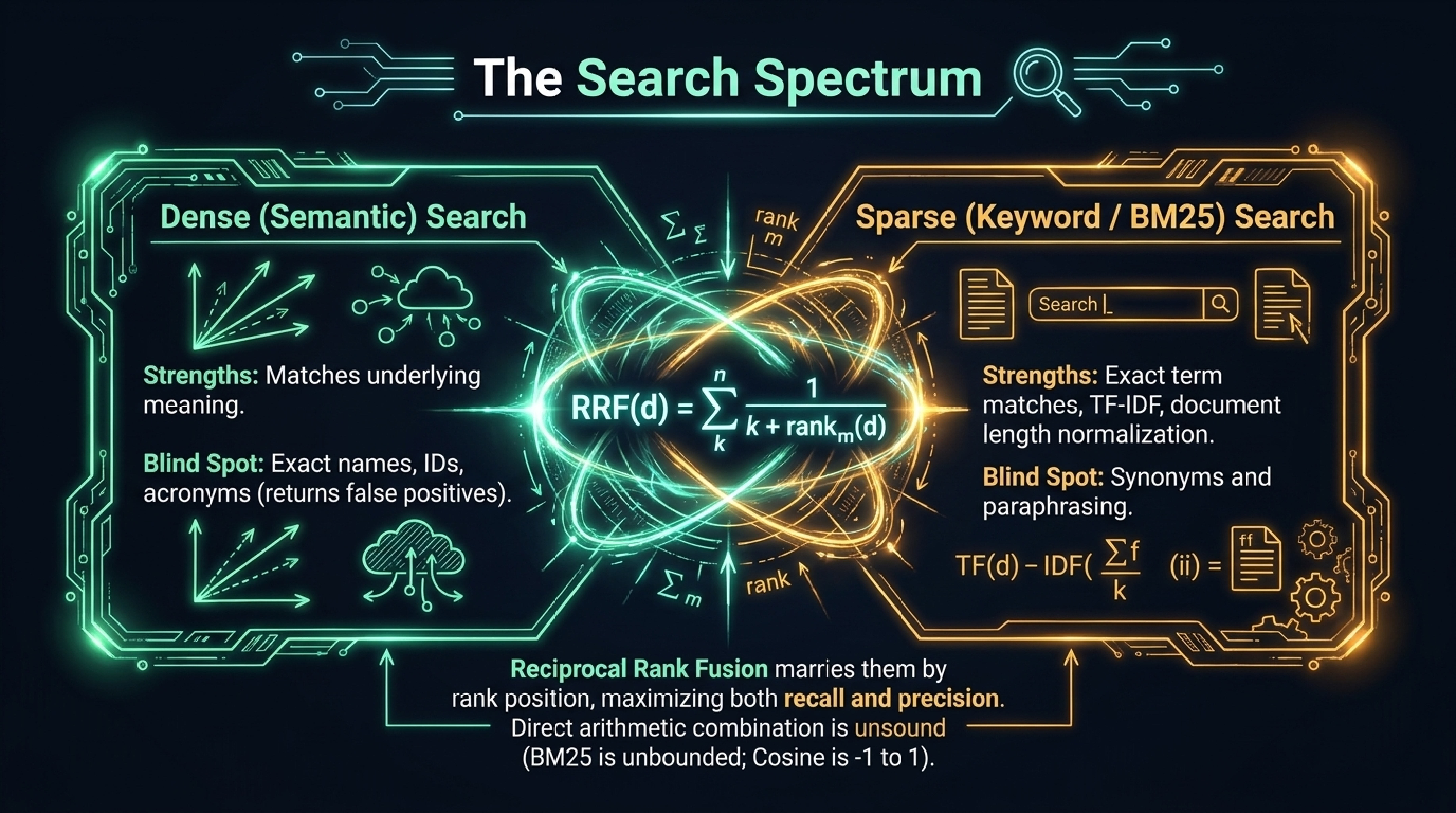


Reranking is the final quality gate. It directly **reduces hallucination** by discarding topically similar but **factually irrelevant chunks** before the LLM ever sees them.



Enterprise Differentiator: Auditability. RAG outputs are uniquely falsifiable because they append document titles and chunk IDs, allowing human verification.

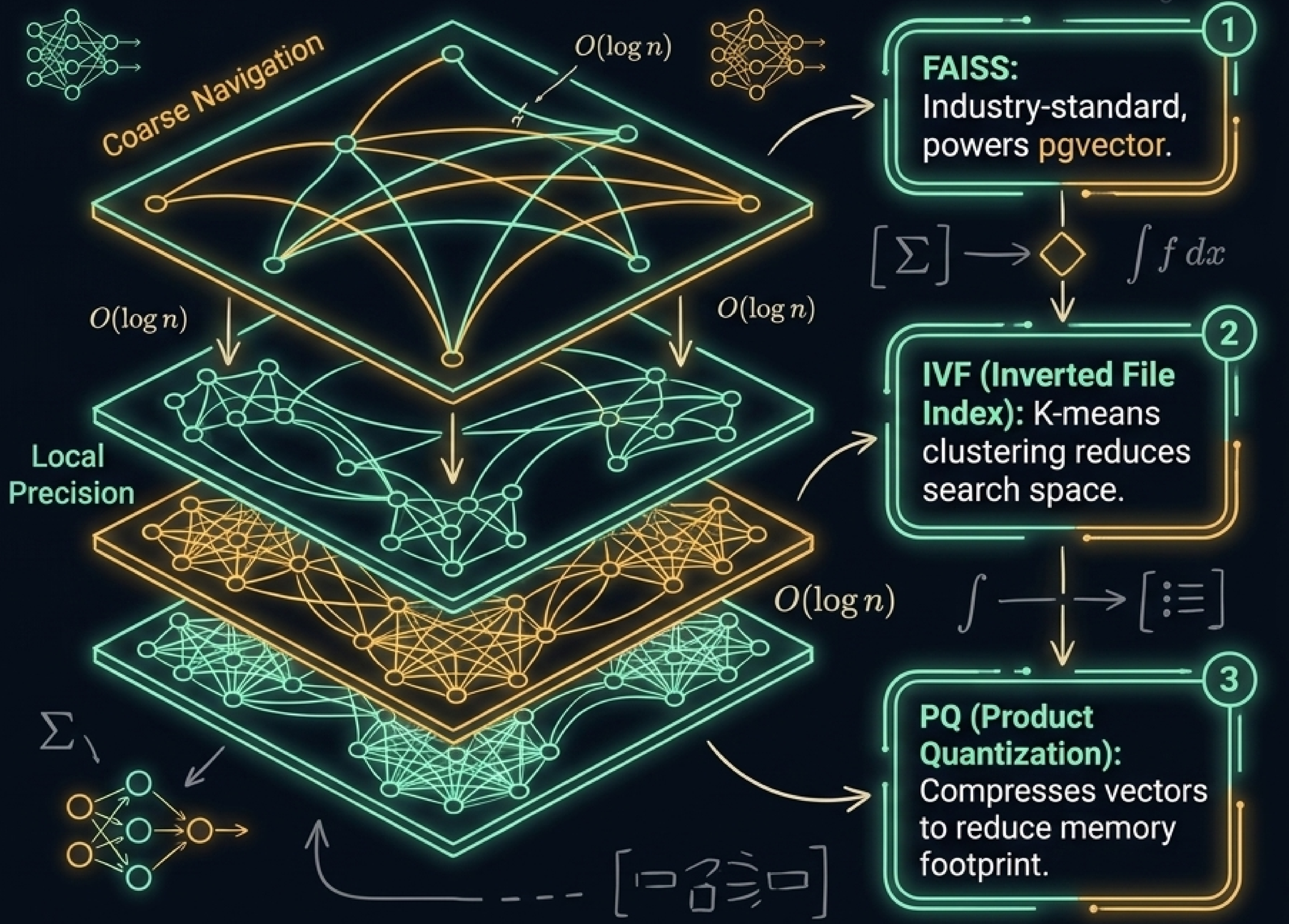

O(log n)
O(log n)
(1, 0, 0,0)
(-0, -1, 0,1)
Dot Product
Magnitude of projection.
Higher positive = more similar.
L2 (Euclidean)
Shortest straight-line distance.
Lower = more similar.
Cosine Similarity
Angle between vectors,
ignoring magnitude. Default
metric for semantic search.
Latent Space: A mathematical construct where
semantically similar concepts are clustered together.
Distance equals relevance.


The Search Spectrum
Dense (Semantic) Search
Strengths: Matches underlying meaning.
Blind Spot: Exact names, IDs, acronyms (returns false positives).
$RRF(d) = \sum_{k}^{n} \frac{1}{k + rank_m(d)}$
Sparse (Keyword / BM25) Search
Search
Strengths: Exact term matches, TF-IDF, document length normalization.
Blind Spot: Synonyms and paraphrasing.
$TF(d) - IDF(\frac{\sum f}{k}$ (ii) =
ff
rank m
rank
$\sum_m$
Reciprocal Rank Fusion marries them by rank position, maximizing both recall and precision.
Direct arithmetic combination is unsound
(BM25 is unbounded; Cosine is -1 to 1).

k-NN (Brute Force): Exact distance, computationally expensive O(n*d). Unscalable.

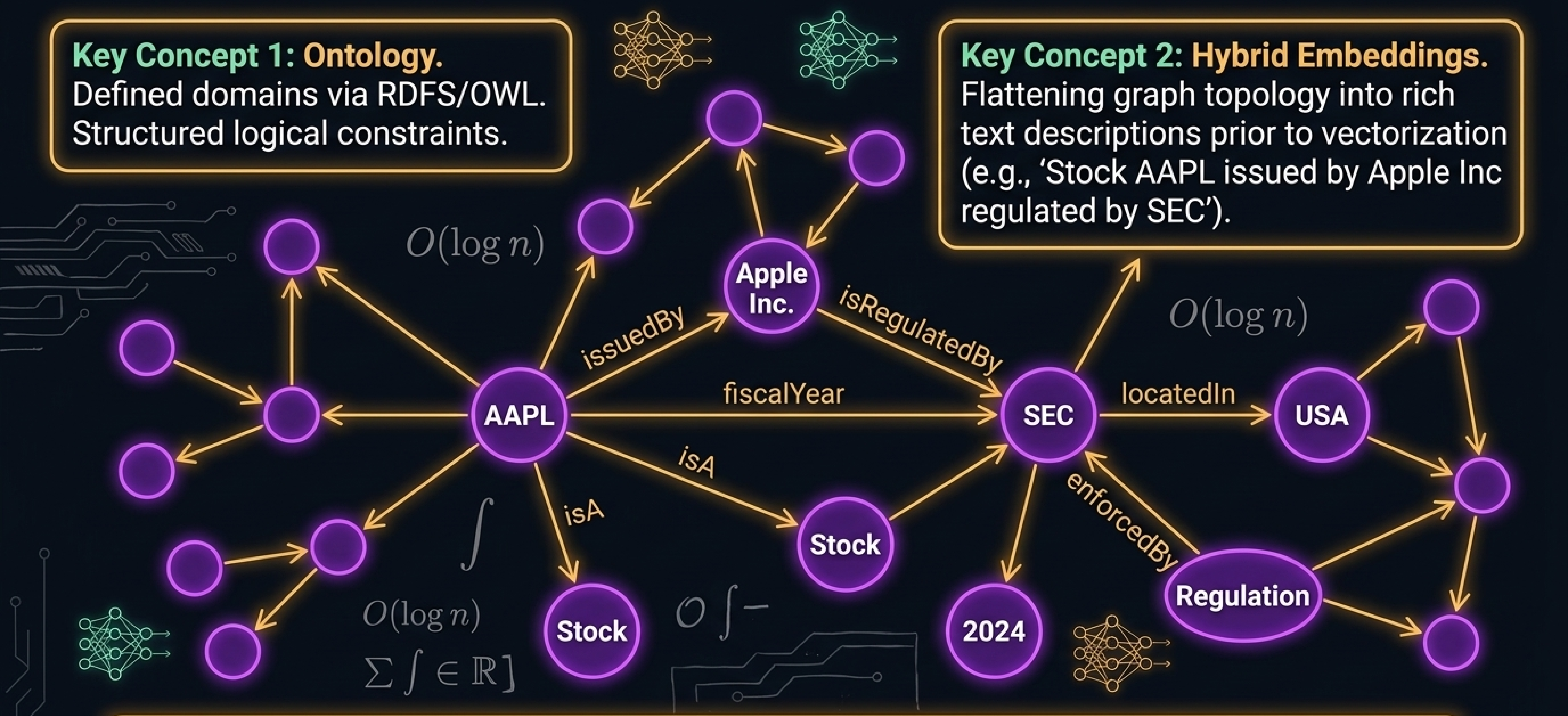
ANN (Approximate Nearest Neighbors): Sacrifices slight accuracy for massive speed O(log n).

HNSW (Hierarchical Navigable Small World)



Key Concept 1: Ontology. Defined domains via RDFS/OWL. Structured logical constraints.

Key Concept 2: Hybrid Embeddings. Flattening graph topology into rich text descriptions prior to vectorization (e.g., ‘Stock AAPL issued by Apple Inc regulated by SEC’).



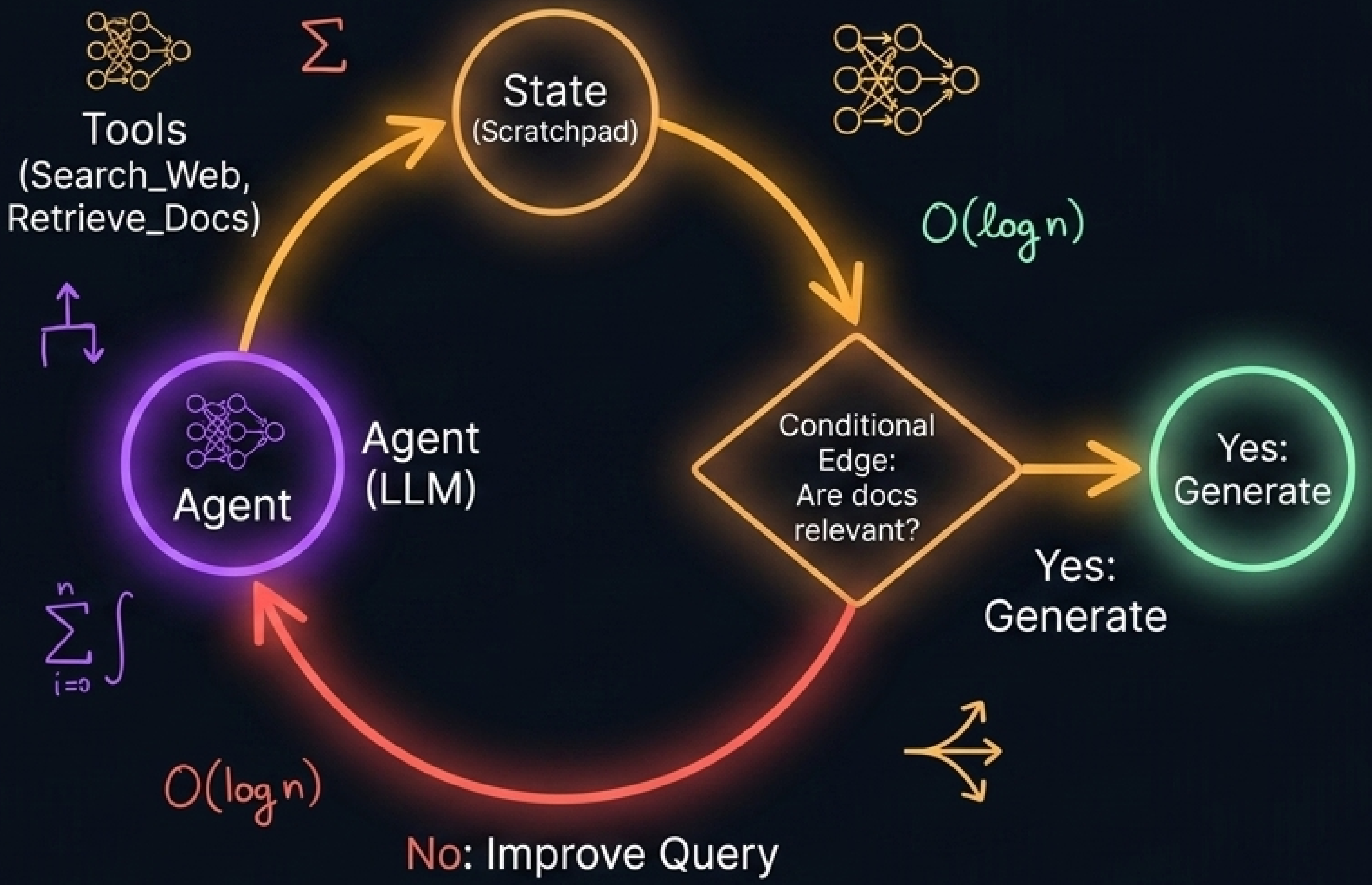
Why Graph RAG? Surpasses flat vector stores by enabling multi-hop reasoning (tracing explicit A → B → C relationships) and guaranteeing traceability via strict schema logic.

Neon Hacker Blueprint

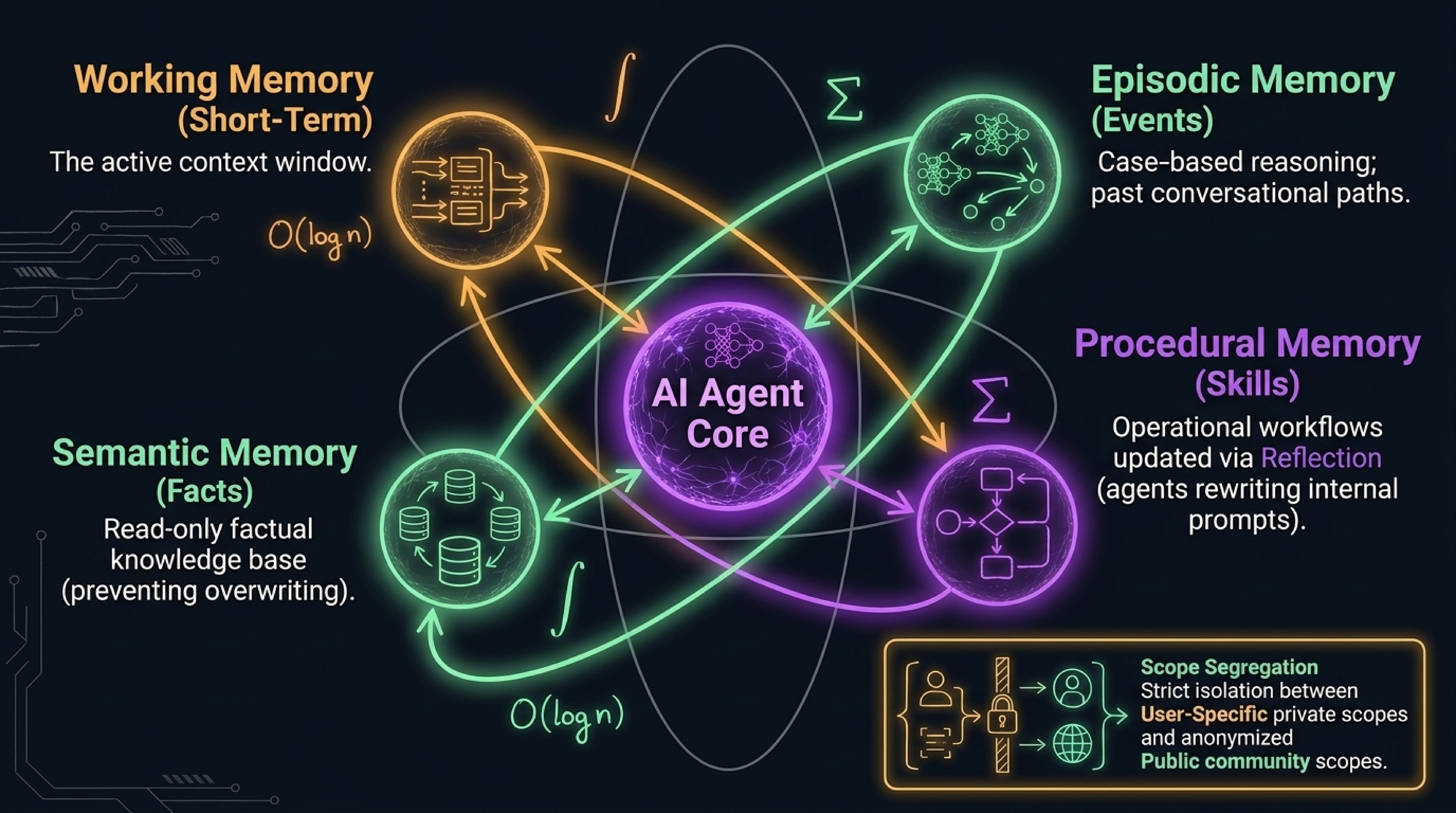
State: The shared active memory/scratchpad.

Nodes: Executable tools or functions.

Conditional Edges: Routing decision points governed by the LLM.

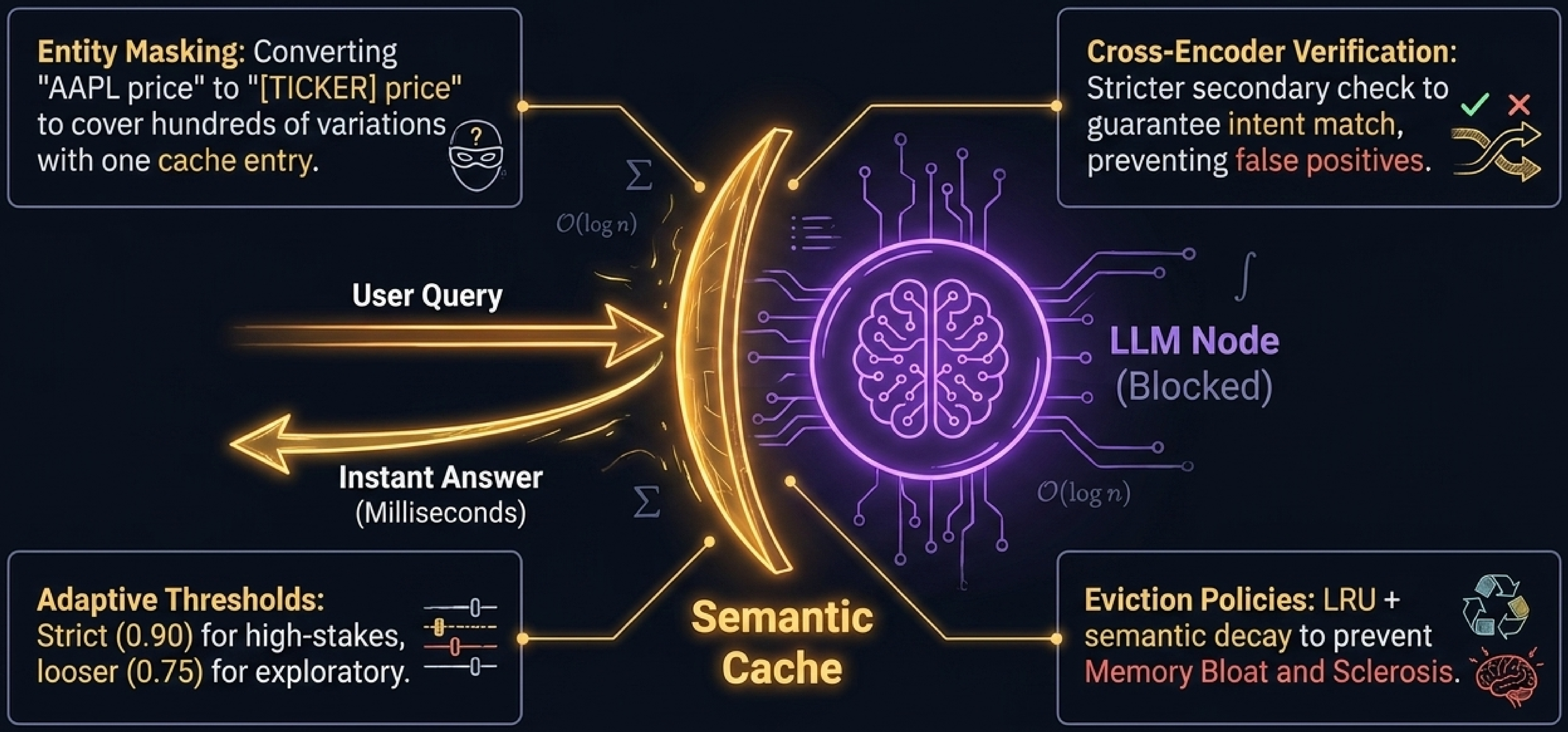


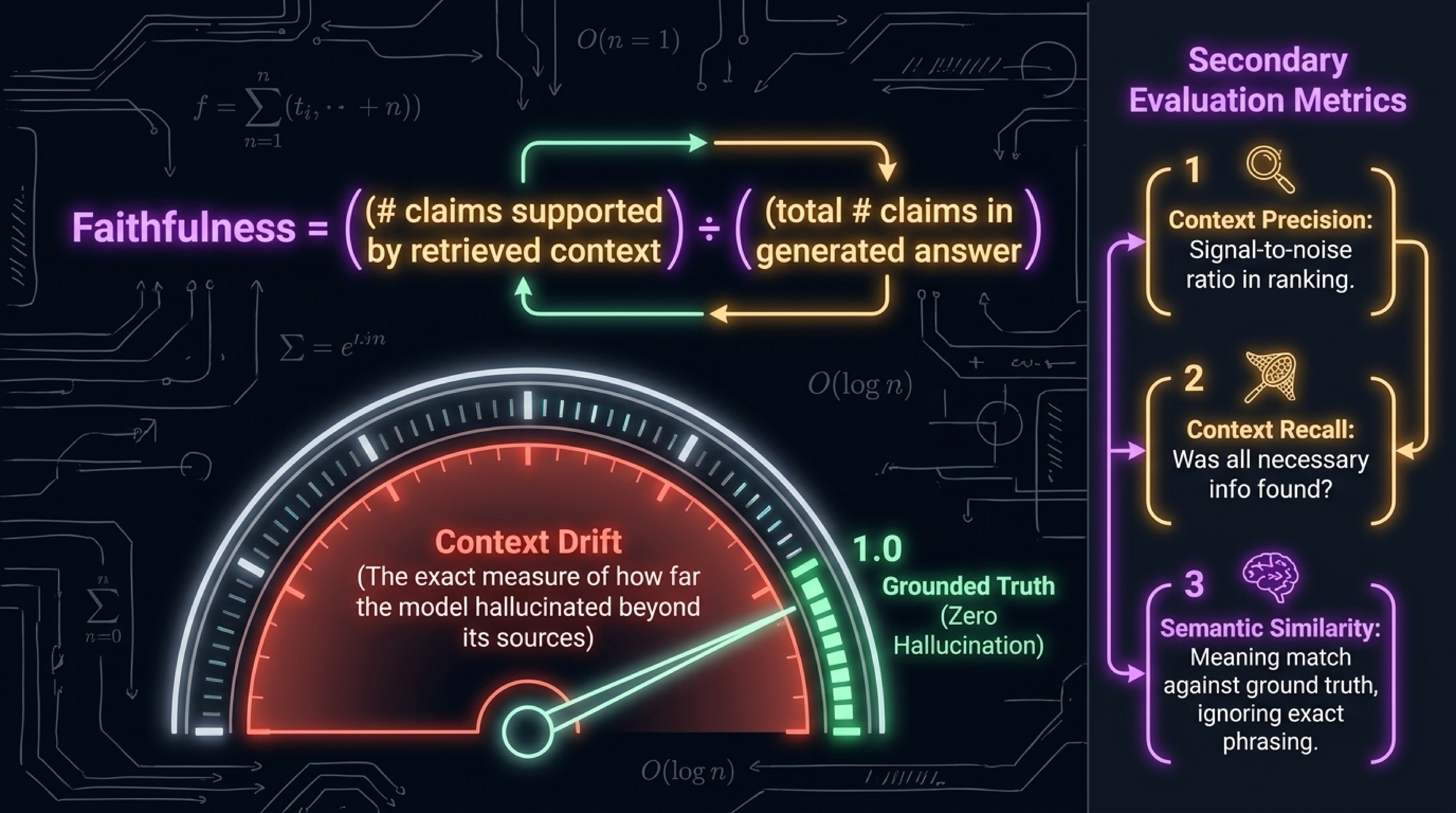
LangGraph Topologies: Moves AI from linear data-fetching to iterative autonomous reasoning loops (**Reason + Act**).



Working Memory
(Short-Term)
The active context window.
O(log n)
Episodic Memory
(Events)
Case-based reasoning; past conversational paths.
AI Agent Core
Procedural Memory
(Skills)
Operational workflows updated via Reflection (agents rewriting internal prompts).
Semantic Memory
(Facts)
Read-only factual knowledge base (preventing overwriting).
O(log n)
Scope Segregation
Strict isolation between User-Specific private scopes and anonymized Public community scopes.

Interceptor Layer: Semantic Cache Shield





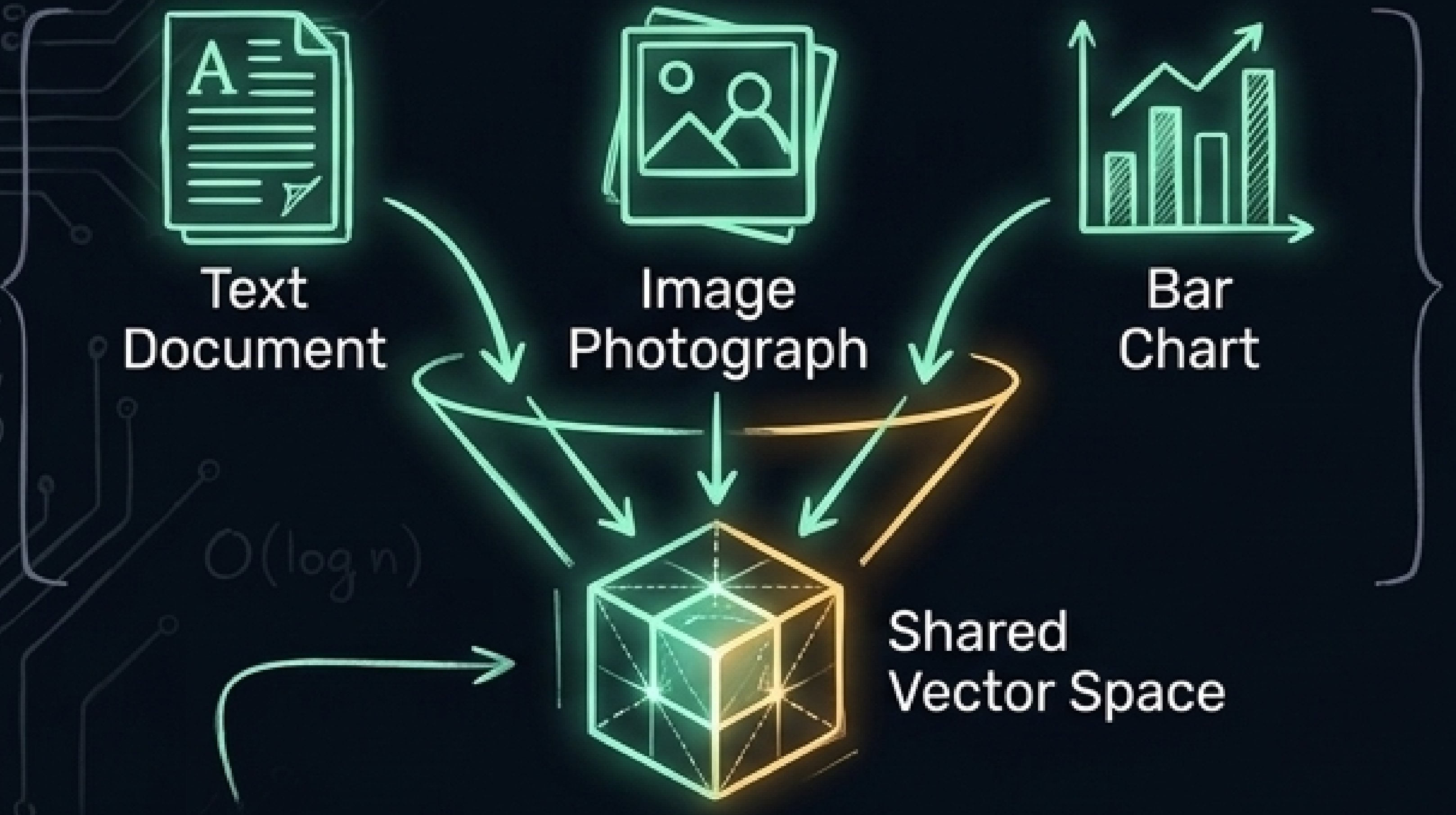
Faithfulness = (# claims supported by retrieved context) ÷ (total # claims in generated answer)
Context Drift
(The exact measure of how far the model hallucinated beyond its sources)
1.0
Grounded Truth
(Zero Hallucination)
Secondary Evaluation Metrics
1
Context Precision:
Signal-to-noise ratio in ranking.
2
Context Recall:
Was all necessary info found?
3
Semantic Similarity:
Meaning match against ground truth, ignoring exact phrasing.

The Production Diagnostic Board

Failure Stage	Symptom	Root Cause	Architect Fix
Retrieval	Finds **wrong** data.	**Vocabulary** mismatch / **Ambiguity**.	**Hybrid Search** + **RRF** / **Query Rewriting** (HyDE).
Context	Missing facts.	**Severed context** from token limits.	**Semantic** & **Hierarchical Chunking**.
Assembly	Correct data ignored.	**Positional bias.**	**Explicit Context Ordering** (first/last placement).
Generation	**Invents** facts.	**Model drifts** beyond context.	**Strict guardrails** + Ragas Faithfulness evaluation.
Systemic	**Cache** gives wrong answer.	**Intent** mismatch.	**Entity Masking** + Cross-Encoder Verification.

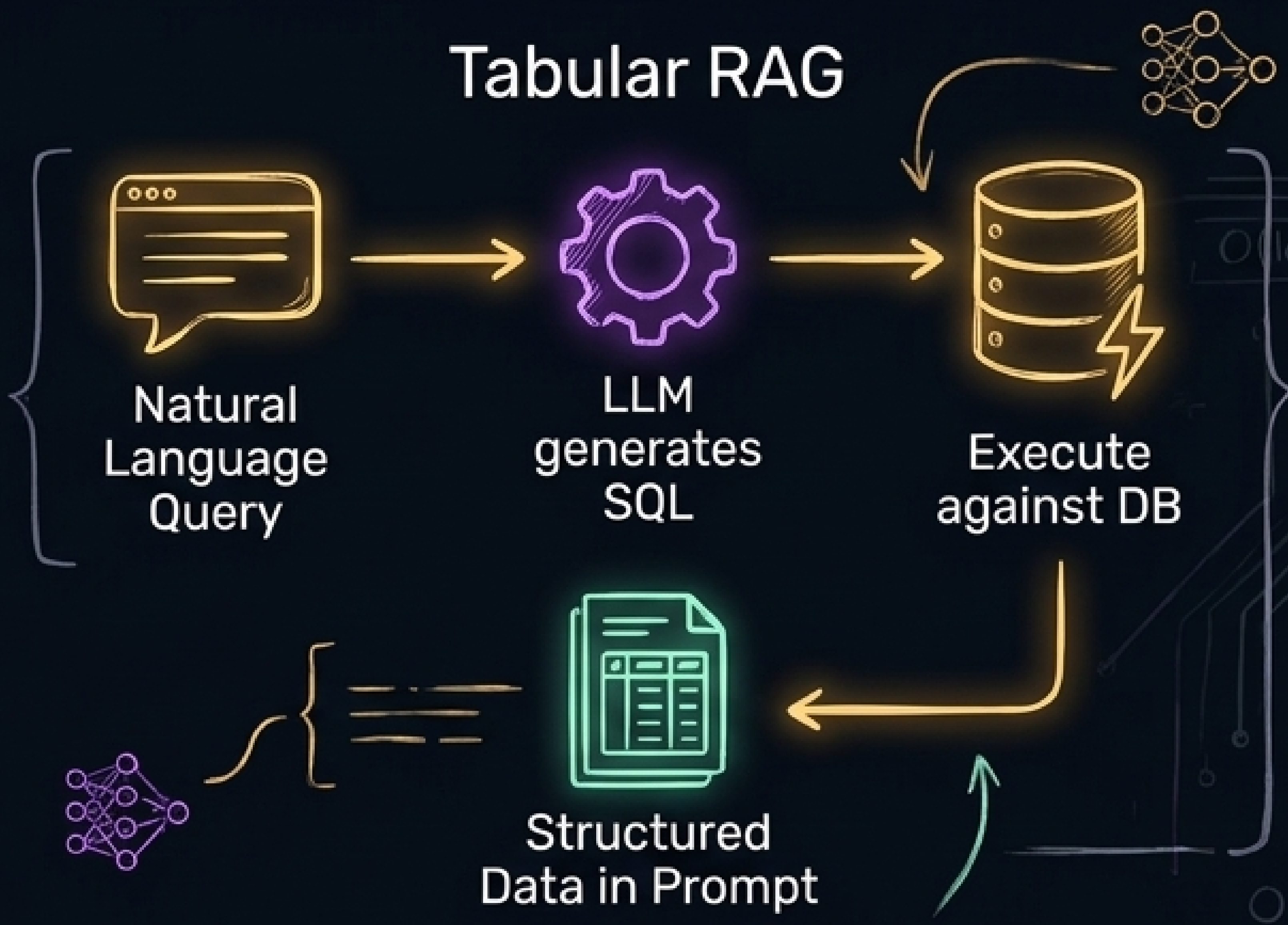
Beyond Flat Text: Multimodal & Tabular RAG Architectures

Multimodal RAG



Using CLIP embeddings to map images and text into identical latent coordinate space, allowing cross-modality retrieval.

Tabular RAG



Injecting live, structured SQL executions dynamically into the generation prompt.

The blueprint remains the same—dynamically binding external reality to generative reasoning.